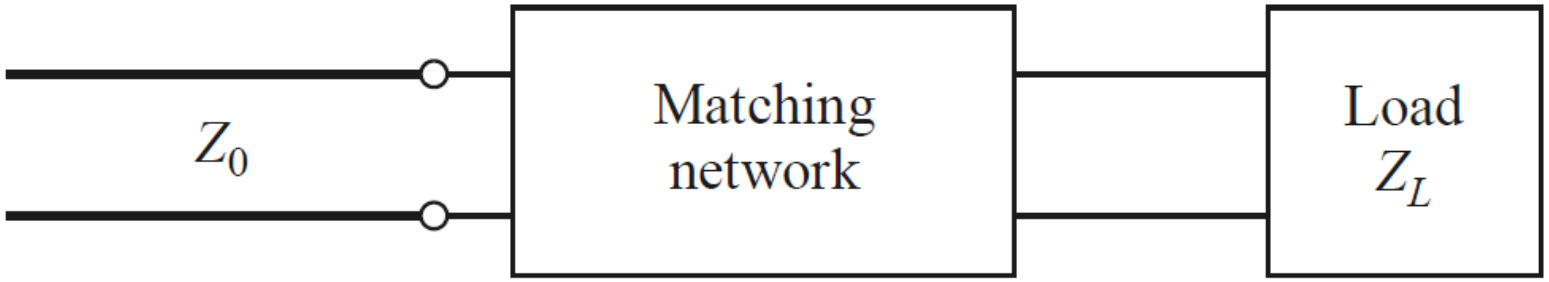
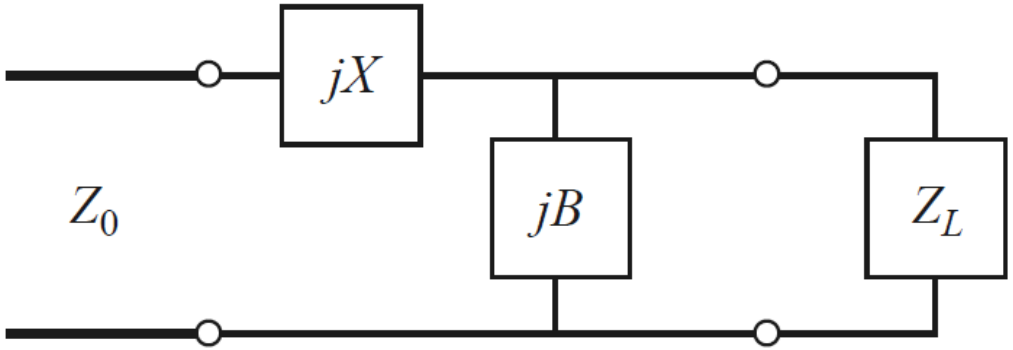
EEME 30004
Communications & Networks

Impedance matching Part 2

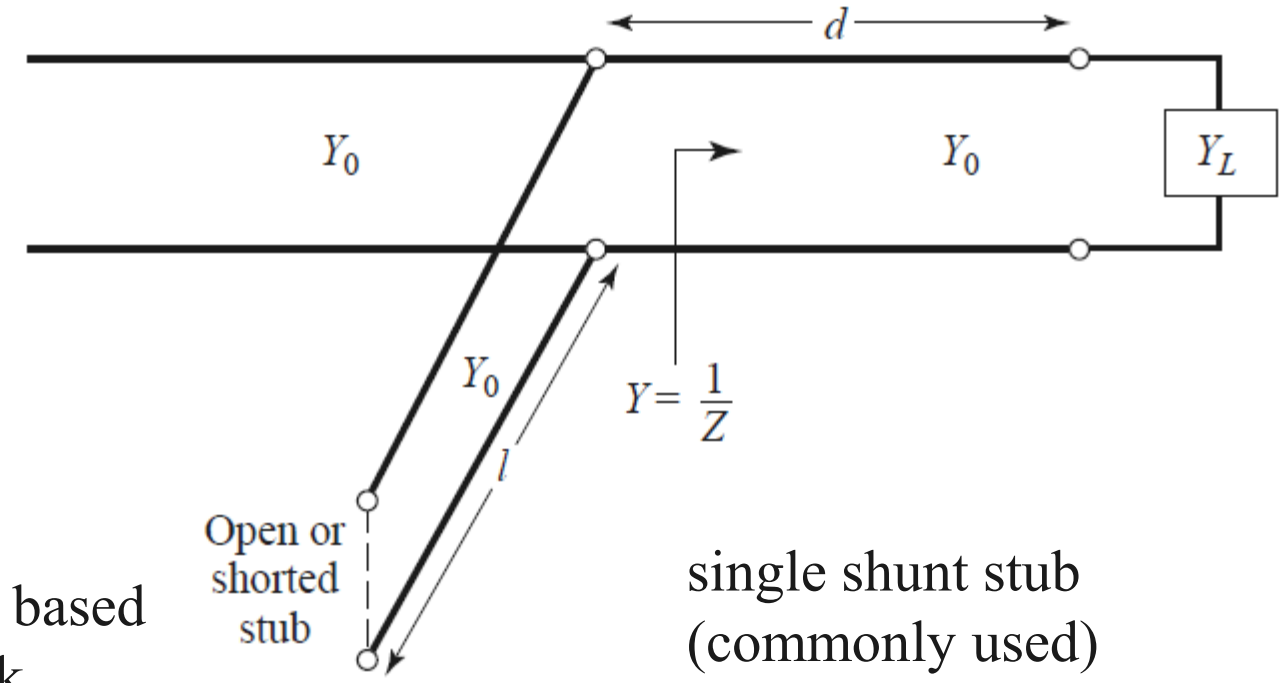
Dr Sean Gu

Impedance matching





LC based matching network

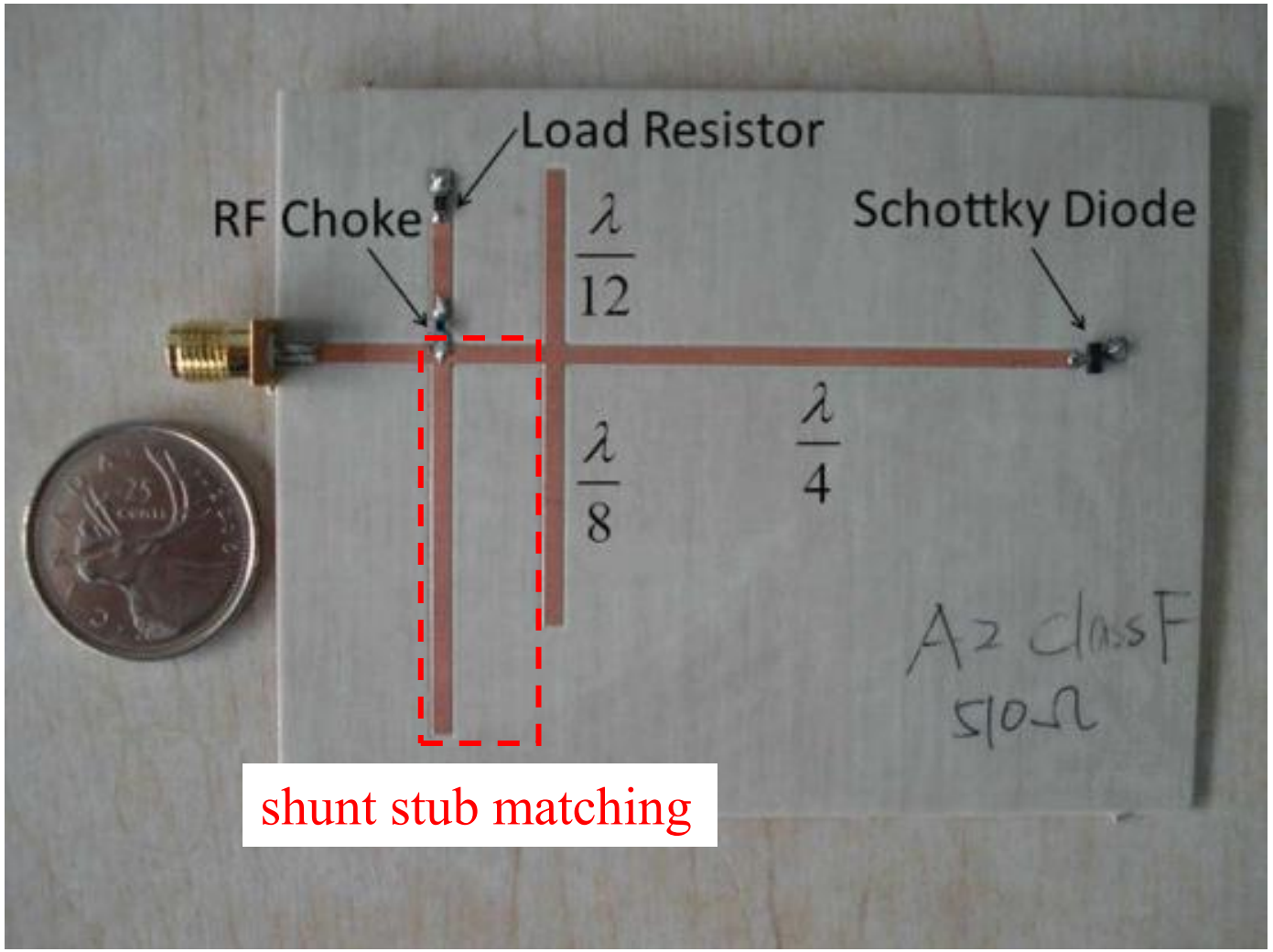


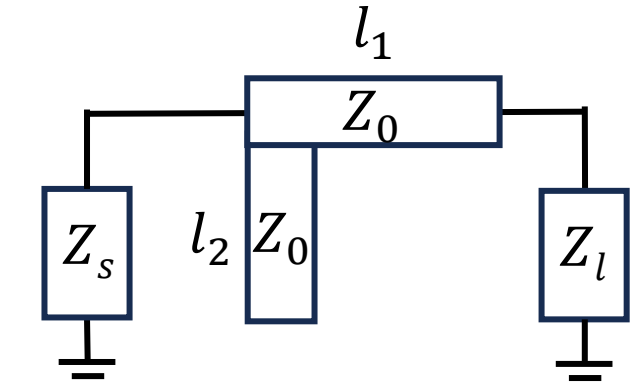
transmission line based matching network

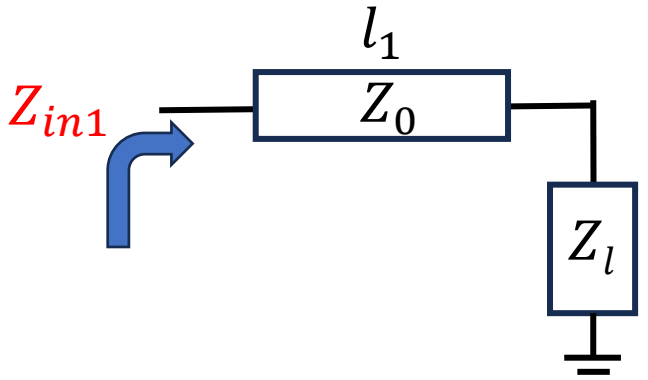
single shunt stub (commonly used)

Impedance matching

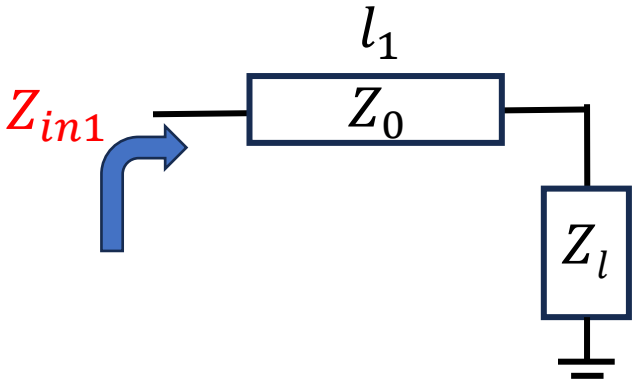
University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering





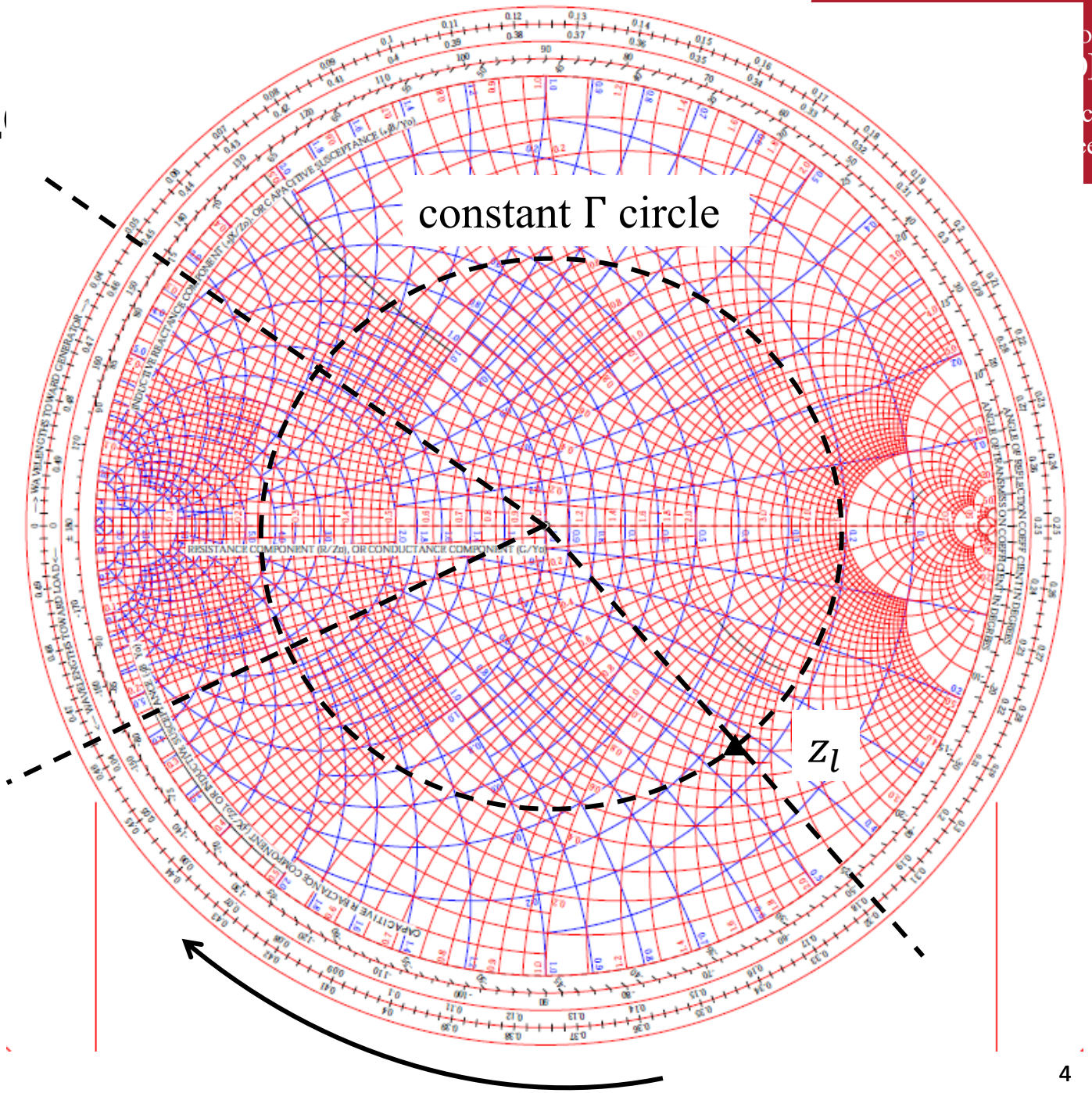


Impedance matchin

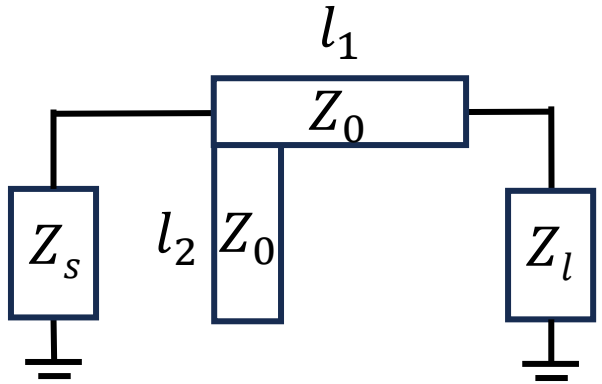


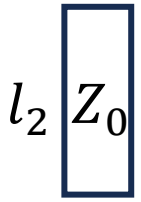


Where do we stop rotating?

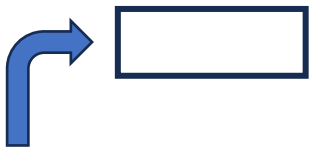


Impedance matchin





stub can be open-circuited or short circuited



open-circuited

$y_{in} = jX$

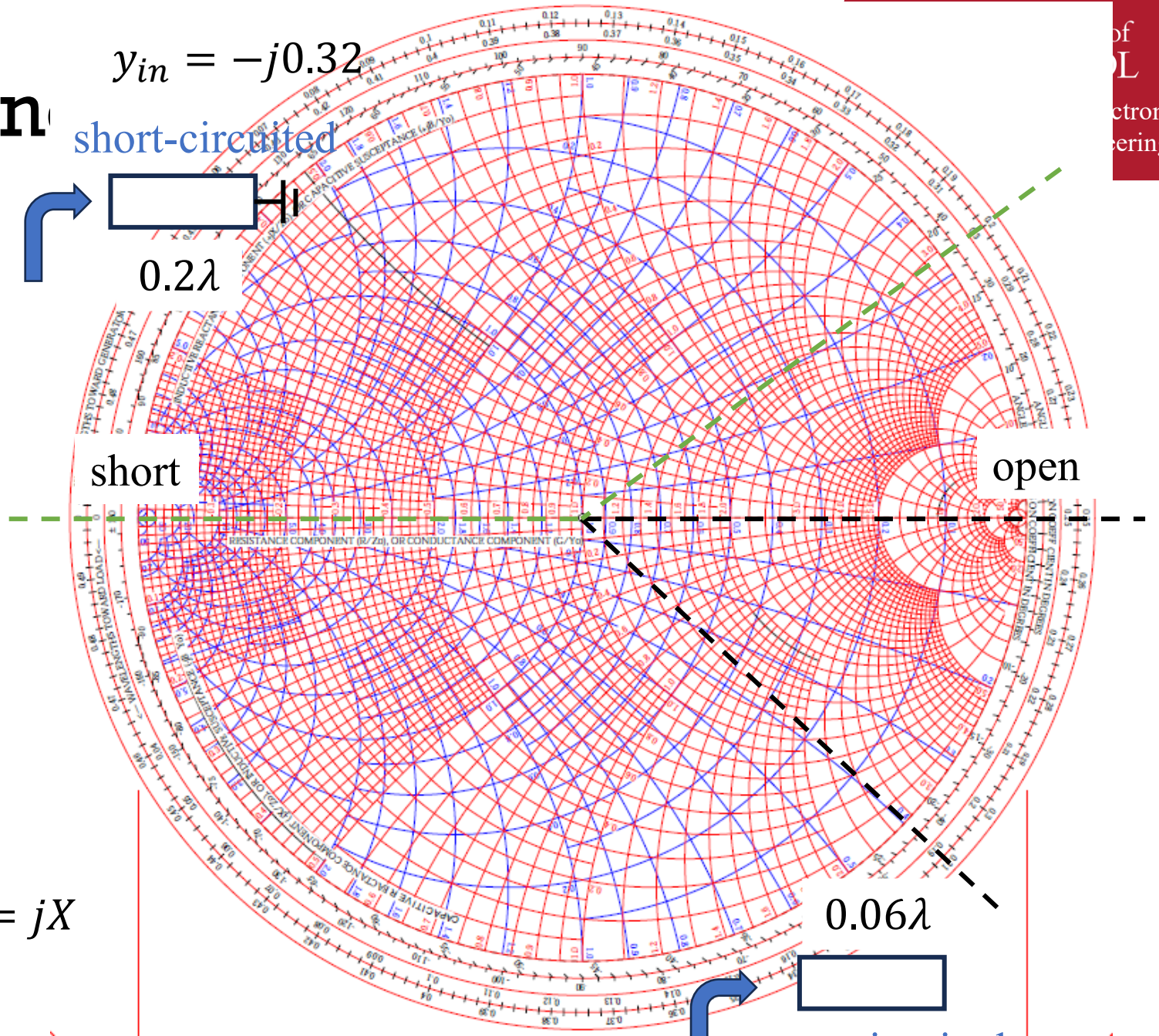


short-circuited

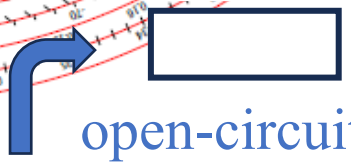
$y_{in} = -j0.32$

short-circuited

0.2λ

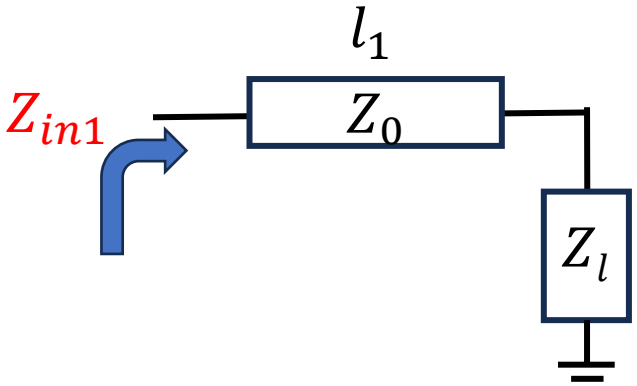


$y_{in} = j0.4$

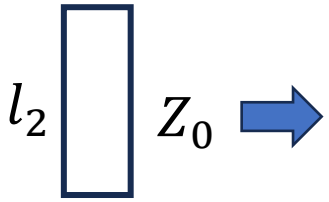


open-circuited

Impedance matchin



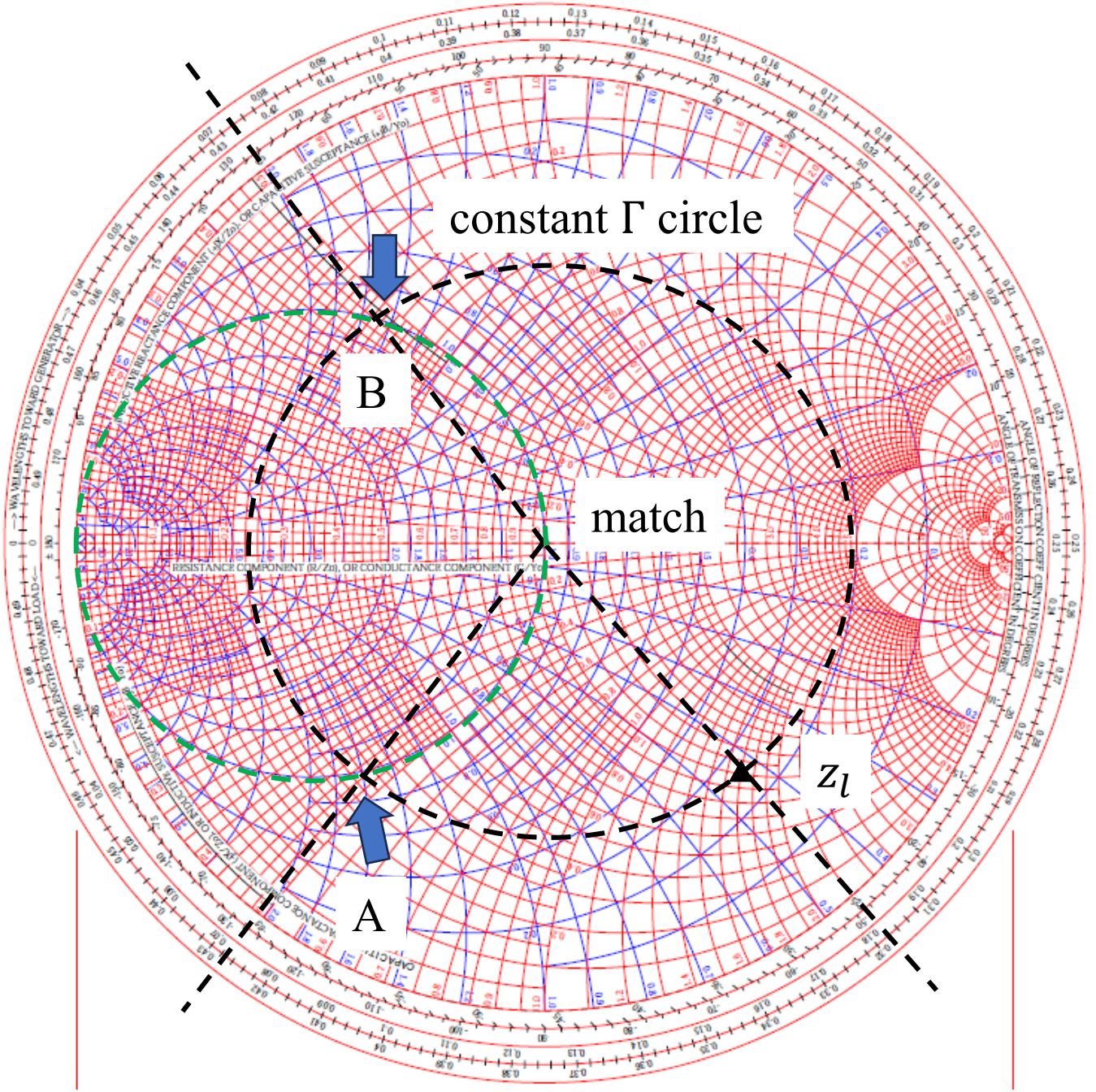
$$y_{in1} = \frac{1}{z_{in1}} = 1 + jX \text{ or } 1 - jX$$



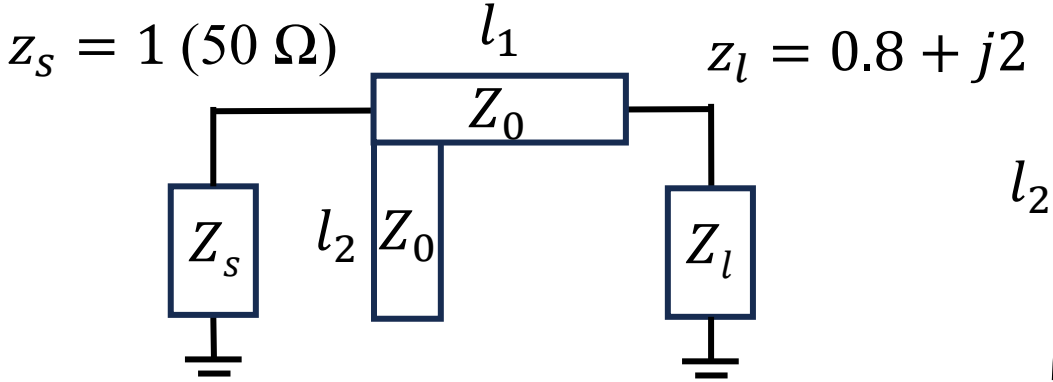
can cancel $\pm jX$

A: $y_{in1} = 1 + jX$

B: $y_{in1} = 1 - jX$



Impedance matchin



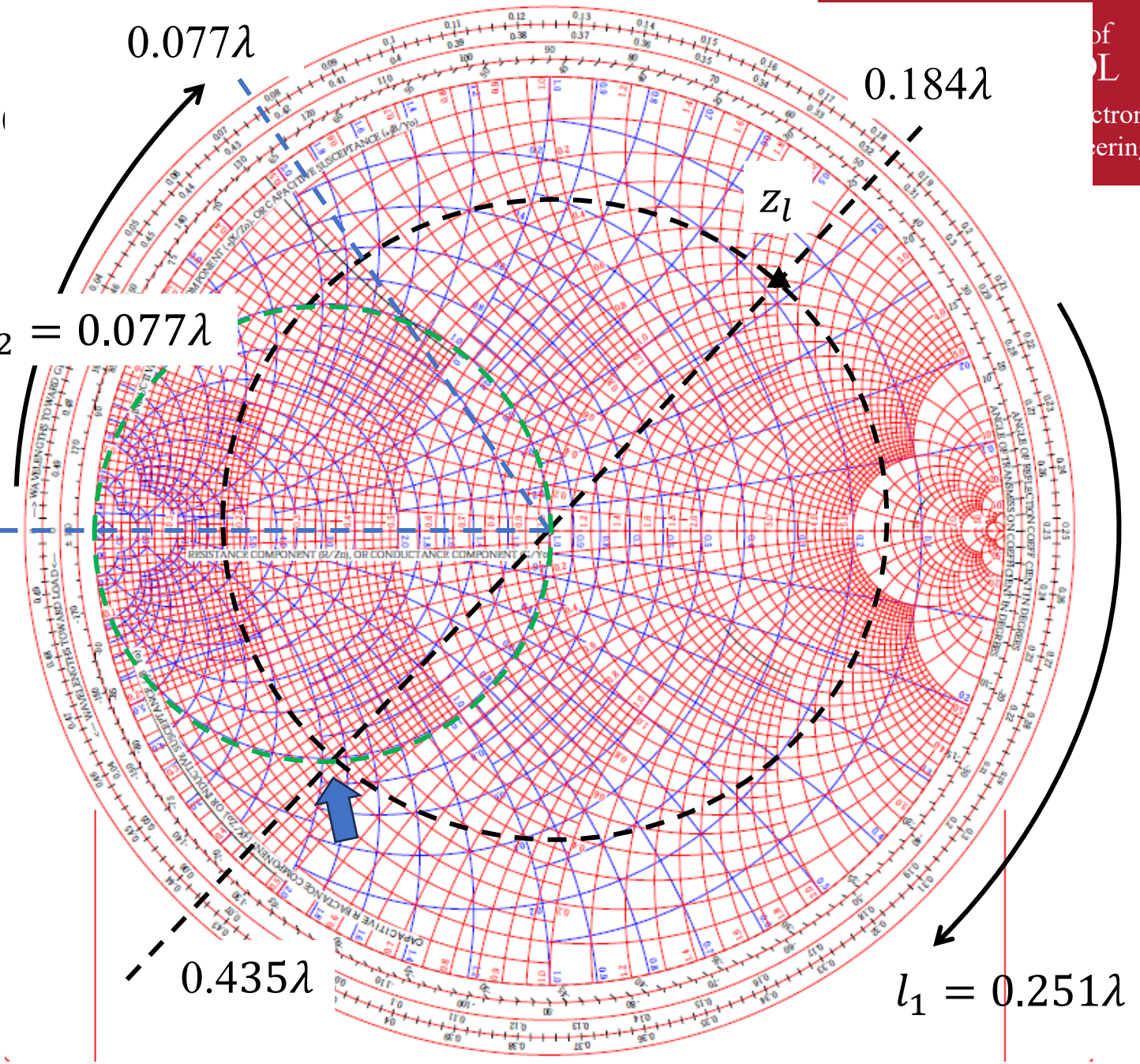
step 1: plot z_l, find initial location

step 2: rotate z_l and find the interception point with $y = 1$ circle, take down location info

step 3: calculate $l_1 = 0.435\lambda - 0.184\lambda = 0.251\lambda$

step 4: read $y_{in} = 1 + j1.9$

step 5: create a shunt stub with $y_{stub} = -j1.9$, short stub (0.077λ)



Questions?

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